

ASS 2 - hamming code

```
#include <stdio.h>
#include <math.h>

int main() {
    int data[20], hamming[30];
    int m, r = 0;

    printf("Enter number of data bits: ");
    scanf("%d", &m);

    printf("Enter data bits:\n");
    for (int i = 1; i <= m; i++)
        scanf("%d", &data[i]);

    while ((pow(2, r)) < (m + r + 1))
        r++;

    int total = m + r;

    int j = 1;
    for (int i = 1; i <= total; i++) {
        if ((i & (i - 1)) == 0)
            hamming[i] = 0;
        else
            hamming[i] = data[j++];
    }

    for (int i = 0; i < r; i++) {
        int pos = pow(2, i);
        int parity = 0;

        for (int j = 1; j <= total; j++) {
            if (j & pos)
                parity ^= hamming[j];
        }

        hamming[pos] = parity;
    }

    printf("\nTransmitted Hamming Code:\n");
    for (int i = total; i >= 1; i--)
        printf("%d ", hamming[i]);
}
```

```

printf("\n");

int received[30];

printf("\nEnter the received Hamming code (position 1 to %d):\n", total);
for (int i = 1; i <= total; i++)
{
    scanf("%d", &received[i]);
}

int error = 0;

for (int i = 0; i < r; i++)
{
    int pos = (int)pow(2, i);
    int parity = 0;

    for (int j = 1; j <= total; j++)
    {
        if (j & pos)
            parity ^= received[j];
    }

    if (parity)
        error += pos;
}

if (error == 0)
{
    printf("\nNo error detected.\n");
}
else
{
    printf("\nError detected at position %d.\n", error);

    received[error] ^= 1;

    printf("Corrected Hamming Code:\n");
    for (int i = total; i >= 1; i--)
        printf("%d ", hamming[i]);

    printf("\n");
}

return 0;
}

```